

# Nina Grant

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/in/nina-grant-2021



## EDUCATION

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- Ph.D. Rutgers University, Atmospheric Science** 2021-2026  
Focus on climate intervention impacts, GPA: 4.0  
Advisors: Alan Robock & Lili Xia
- M.S. Rutgers University, Atmospheric Science** 2023  
Focus on agricultural impacts of climate intervention, GPA: 4.0  
Advisors: Alan Robock & Lili Xia
- B.A. Princeton University, Geosciences** 2021  
Focus on climate science and sustainability  
Advisor: Gabriel Vecchi

## RESEARCH EXPERIENCE

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**Senior Computational Climate Scientist** Apr 2026 – Present  
interos.ai Remote

Supervisor: Andrea Little Limbago

- Developing and validating computational ESG, climate, and catastrophic risk models, translating them into decision-relevant insights for global supply chains.

**Research Assistant** Aug 2021 – Apr 2026  
Rutgers Impact Studies of Climate Intervention Lab New Brunswick, NJ  
Advisors: Dr. Alan Robock and Lili Xia

***Tropical Agriculture Responses to Stratospheric Aerosol Intervention***

- Investigated how stratospheric aerosol injections (SAI) impact Indian wheat and rice crop suitability and production using CESM2 ARISE-SAI-1.5 model outputs
- Researching the impact of climate intervention and nuclear war on coffee and cocoa yields in Brazil and Ghana using multi-model downscaled ISIMIP data
- Compared the skill and suitability of machine learning and statistical downscaling methods for SAI scenarios

**Weather & Climate Data Analyst Summer Student Employee** Jun 2023 – Jan 2024  
Electric Power Research Institute Remote  
Tech Lead: Erik Smith

- Quantified the skill of 4 gridded datasets (ERA5, ERA5-Land, MERRA-2, and PRISM) at representing extreme temperatures and precipitation over the conterminous U.S.
- Processed 40+ years of high-resolution climate data using Python (xarray, Dask, Rasterio) to evaluate dataset biases across temporal and spatial scales.

**Senior Thesis**

Princeton University

Advisor: Dr. Gabriel Vecchi

Aug 2020- May 2021

Princeton, NJ

***Solar Geoengineering Effects on the Indian Monsoon Precipitation Patterns***

- Analyzed GFDL climate model simulations of hemispherically varying volcanic eruptions and a 1% decline in solar radiation to isolate the impact of injection location and amount on monsoonal behavior
- Investigated whether local thermodynamics or large-scale dynamics were responsible for the pattern shifts

**Internship**

Princeton Climate Action Plan Emission Reduction Strategies Team

Project Lead: Nic Choquette-Levy

Dec 2020- May 2021

Princeton, NJ

- Researched decarbonization transitions for Princeton landscaping services by deducing current GHG emissions and sound pollution to provide recommendations to the municipality of Princeton
- Served as translator during landscaper interviews and translated documents to Spanish to disseminate team findings to local communities

**Junior Independent Work**

Princeton University

Advisor: Dr. Gabriel Vecchi

Sep 2019- Jan 2020

Princeton, NJ

***Sea Surface Temperature Variability as A Predictor of Indian Monsoon Activity***

- Revisited past work by Prof. Vecchi to verify whether, with more data, a predictive relationship persisted between breaks in the monsoon and sea surface temperatures in the Indian Ocean and the Bay of Bengal

**Research Internship**

ALMA Observatory

Advisors: Chentao Yang and Violette Impellizzeri

Jun 2019 – Aug 2019

Santiago, Chile

- Applied a new imaging technique for carbon tracing in deep-space, red-shifted galaxies acquiring experience with Linux OS, virtual machines, data analysis, presenting scientific research, and bilingual communication skills

**PROFESSIONAL TRAINING**

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**Young Voices of Science Workshop**

Hubbard Brook Research Foundation (Online)

May 2026

- Eight-week training program led by expert science communication consultants designed to equip undergraduate and graduate students in environmental fields with effective methods for sharing their scientific work and passions with non-technical audiences.
- Participants also complete at least one outreach project or activity, putting what they learn into practice.

**Leadership & No-Blame Problem Solving Certificate** Jun 2025  
Rutgers University (Online)

- Asynchronous 10-module course on civic leadership and practical governance skills using the No-Blame Problem Solving method. Taught how average citizens can enact local change. Covered citizen rights, local power structures, media literacy, and strategies for engaging in government decision-making.

**AGU Local Science Partner Spring Congressional Visit Workshop** Mar 2025  
Washington, D.C.

- Full-day workshop on science policy engagement, science communication, media advocacy, and congressional visit preparation. Included training on op-ed writing, messaging strategy, and mock meetings with legislators.

**Climate Change AI Virtual Summer School** Aug 2024  
Online

- 3-month course on the applications of AI and machine learning for tackling climate change problems with lectures and hands-on coding tutorials. Topics spanned agriculture, biodiversity, climate science, GHG accounting and monitoring, power systems, transportation, and more.

**SciComp Workshop - Collaborative Coding with GitHub** Dec 2023  
AGU Fall Meeting 2023, San Francisco, CA

- 4-hr workshop on version control, git, and GitHub best practices and how to use GitHub for scientific research projects

**Machine Learning Specialization** Sep 2023  
Online

- 10-week course hosted on Coursera by Stanford University and DeepLearning.AI on supervised learning (linear regression, logistic regression, neural networks, decision trees), unsupervised learning (clustering, anomaly detection), recommender systems, and reinforcement learning

**CESM Tutorial** Jul 2023  
Boulder, Colorado

- Selective week-long intensive on the science and advancements behind the CESM climate model and practical training exercises on running custom model simulations

**CESM AGU Workshop** Dec 2022  
AGU Fall Meeting 2022, Chicago, IL

- 4-hr workshop on running simple custom models in CESM on the AWS cloud

## **LEADERSHIP (VOLUNTEER/SERVICE)**

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Solar Geo Society Ambassador 2024-Present

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|-----------------------------------------------------------------------------------|--------------|
| AGU Local Science Partner                                                         | 2024-Present |
| Organizer of the 1 <sup>st</sup> Rutgers Atmospheric Science Grad Student Retreat | 2024         |
| Student Convener of AGU'24 Session GC009                                          | 2024         |
| Co-Chair of the 2026 Gordon Research Seminar                                      | 2024-Present |
| Conference Coordinator of the Princeton Energy Association Fall Conference        | 2020         |
| <b>Reviewer:</b> Earth's Future (2025)                                            |              |

## HONORS AND AWARDS

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| • <b>Sigma Xi Student Research Showcase – 1<sup>st</sup> Place Graduate Division</b> (\$500) | 2026 |
| • Young Voices of Science Fellow                                                             | 2026 |
| • Sigma Xi Nomination                                                                        | 2025 |
| • Atmospheric Science Graduate Student Travel Grant (\$2000)                                 | 2025 |
| • Solar Geo Society Ambassador                                                               | 2024 |
| • Atmospheric Science Graduate Student Travel Grant (\$1200)                                 | 2024 |
| • <b>Elected Co-Chair of the 2026 Gordon Research Seminar</b>                                | 2024 |
| • Atmospheric Science Graduate Student Travel Grant (\$1500)                                 | 2023 |
| • RCEI Travel Award (\$500)                                                                  | 2023 |
| • RCI Fund Travel Award (\$250)                                                              | 2022 |
| • AGU Student Travel Grant (\$1000)                                                          | 2022 |
| • <b>Dean's Fellowship, Rutgers University School of Graduate Studies</b> (\$49,736)         | 2021 |

## PUBLICATIONS

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[Google Scholar](#) | **ORCID: 0000-0003-4188-9275**

**Grant, N.**, Robock, A., Xia, L., Kiniry, J., and Clark, B. Can Solar Climate Intervention Save Coffee and Chocolate from Climate Change? (Under Review)

**Grant, N.**, Kiniry, J., & Aziz, F. (2025) Modeling the impacts of climate change on cocoa. In *Theobroma cacao - Past, Present and Future Insights*. IntechOpen.  
<https://doi.org/10.5772/intechopen.1011776>

Smith, E., **Grant, N.**, Luo, X. et al. (2025). Evaluating the ability of gridded climate datasets to capture temperature and precipitation trends and extremes. *Sci Rep*, 15(1), 12607.  
<https://doi.org/10.1038/s41598-025-97570-7>

**Grant, N.,** Robock, A., Xia, L., Singh, J., and Clark, B. (2025). Impacts on Indian agriculture due to stratospheric aerosol intervention using agroclimatic indices. *Earth's Future*, 13(1), e2024EF005262. <https://doi.org/10.1029/2024EF005262>.

Singh, J., Sahany, S., Xia, L., Robock, A., and **Grant, N.** Comparing quantile mapping and other statistical methods in downscaling rainfall for agriculture impacts. (Submitted)

## INVITED TALKS

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| <b>Invited Talk,</b> <i>A brief introduction to the physical science of SRM</i> , New York, NY                                                                                                  | Sep 2025 |
| <b>Invited Talk,</b> <i>The Modern Challenges of Geoengineering</i> , Global Affairs Canada, Virtual                                                                                            | Jul 2025 |
| <b>Invited Discussion Leader,</b> <i>Climate Engineering: Processes, Uncertainties, Responses, and Impacts</i> , Gordon Research Conference on Climate Engineering, Lucca (Barga), Lucca, Italy | Feb 2024 |
| <b>Invited Talk,</b> <i>How I Got Involved in Sustainability Work</i> , The Hewitt School, Virtual                                                                                              | May 2021 |

## CONFERENCE PRESENTATIONS

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| <b>Oral Presentation,</b> <i>Keeping Coffee and Chocolate on the Table in a Warming World</i> , AGU Fall Meeting, New Orleans, LA, USA                                                                                      | Dec 2025 |
| <b>Poster Presentation,</b> <a href="#"><i>Comparing Machine Learning and Traditional Downscaling Methods for Climate Projections Under Stratospheric Aerosol Intervention</i></a> , AGU Fall Meeting, New Orleans, LA, USA | Dec 2025 |
| <b>Oral Presentation,</b> <i>Can stratospheric aerosol intervention save coffee and chocolate from climate change?</i> Degrees Global Forum, Cape Town, South Africa                                                        | May 2025 |
| <b>Poster Presentation,</b> <a href="#"><i>Impacts on Indian Agriculture Due to Solar Climate Intervention Using Agroclimatic Indices</i></a> , Degrees Global Forum, Cape Town, South Africa                               | May 2025 |
| <b>Poster Presentation,</b> <a href="#"><i>Can stratospheric aerosol intervention save coffee and chocolate from climate change?</i></a> AGU Fall Meeting, Washington, D.C., USA                                            | Dec 2024 |
| <b>Poster Presentation,</b> <i>Enhancing Climate Forcing Data for Crop Models: Addressing Challenges in Downscaling and Bias Correction under Climate</i>                                                                   | Dec 2024 |

*Intervention and Nuclear Winter Scenarios*, AGU Fall Meeting, Washington, D.C., USA

**Poster Presentation**, [\*Can Solar Climate Intervention Save Coffee and Chocolate from Climate Change?\*](#) 14<sup>th</sup> GeoMIP Meeting, Ithaca, NY, USA Jul 2024

**Poster Presentation**, [\*Impacts on Indian Agriculture Due to Solar Climate Intervention Using Agroclimatic Indices\*](#), Gordon Research Conference on Climate Engineering, Lucca (Barga), Lucca, Italy Feb 2024

**Poster Presentation**, [\*Impacts on Indian Agriculture Due to Solar Climate Intervention Using Crop Suitability and Agroclimatic Indices\*](#), AGU Fall Meeting, San Francisco, CA, USA Dec 2023

**Poster Presentation**, *Assessing the Ability of Reanalysis Data to Identify Local Extremes*, AGU Fall Meeting, San Francisco, CA, USA Dec 2023

**Poster Presentation**, [\*Indian Agricultural Impacts under GeoMIP G6 Experiments\*](#), AGU Fall Meeting, Chicago, IL, USA Dec 2022

**Poster Presentation**, [\*Indian Monsoon Precipitation Changes in CESM2-WACCM GeoMIP6 Experiments\*](#), Gordon Research Conference on Climate Engineering, Newry, ME, USA Jun 2022

## TEACHING EXPERIENCE

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**Guest lecturer – Climate Modeling** Fall 2024

Rutgers University (Remote)

16:107:544 Modeling of Climatic Change (Graduate)

- Led unit on a Python-based global climate model, deployed and tested model on Rutgers's Amarel HPC, created reproducible setup via GitHub, and developed assignments, tutorials, and exercises. Gave guest lecture, supported students through the assignment, and coordinated with the model developer.

**Tutor - Math** Feb-Jul 2024

Private (Remote)

- Tutored a high school student in algebra and geometry, raising GPA by 1.6 points

**Tutor - Math** Sep 2022 – Apr 2023

Rutgers School of Environmental and Biological Sciences

- Tutored 14 undergraduate students in college-level mathematics (geometry, algebra, calculus) in weekly group and individual sessions, developed lesson plans, coordinated schedules for virtual and in-person sessions, adapted to the needs of the students each session

## PROFESSIONAL AFFILIATIONS

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|------------------------------------------------------------|--------------|
| Sigma Xi                                                   | 2025-Present |
| National Association of Black Geoscientists (NABG)         | 2025-Present |
| American Association for the Advancement of Science (AAAS) | 2024-Present |
| American Geophysical Union (AGU)                           | 2022-Present |
| National Society of Black Engineers (NSBE)                 | 2018-2021    |

## SKILLS AND TOOLS

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**Programming languages:** Python, Bash script, MATLAB, SQL, HTML

**Python Packages:** Xarray, Numpy, Pandas, Matplotlib, Cartopy, Salem, Seaborn, PyTorch, Scikit-learn, Keras, SciPy, PyAutoGUI

**Applications and Platforms:** VSCode, GitHub, Snowflake, Spyder IDE, MobaXTerm, DaVinci Resolve, HPCs (Cheyenne/Derecho, Amarel, Frontera)

**Languages:** English (native), Spanish (near native)